Pratyush Patra

Master in Statistics
Indian Statistical Institute, Kolkata

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Degree/Course	Institution	Year	Aggregate %
M.Stat	Indian Statistical Institute, Kolkata	2026–2028	NA
B.Stat (Hons.)	Indian Statistical Institute, Kolkata	2023–2026	**88.61%**
Higher Secondary (+2)	Baranagar R.K. Mission High School	2021–2023	95.0%
Secondary (10th)	Baranagar R.K. Mission High School	2011–2021	90.0%

ACADEMIC PROFILE & INTERESTS

Currently a M.Stat student at ISI Kolkata, I specialize in the intersection of Analyzing Health Data, Longitudinal Data Analysis and Machine Learning. My primary research interests lie in developing Causal Inference frameworks and machine learning tools for observational biomedical data, focusing on longitudinal treatment regimes, confounder selection in high dimensions, and biomarker discovery in precision medicine.

SCHOLASTIC ACHIEVEMENTS

- **Rank 36** in ISI Admission Test (National Level) among thousands of applicants [2023]
- **3rd Rank** in B.Stat batch.
- **Rank 125** in West Bengal Joint Entrance Examination (WBJEE) [2023]
- **Rank 5187** in JEE Advanced; Selected for the **Indian Institute of Science (IISc)** [2023]

RESEARCH EXPERIENCE

- **Big Data Summer Immersion at Yale 2026 | Yale School of Public Health** [Jun 2026 - Jul 2026]
  - Performed **Longitudinal Data Analysis** on **MIMIC-IV** EHR data as a part of the **Electronic Health Records** research group of BDSY 2026.
  - Tackled challenges of **Within Subject Correlation**, **Variable Selection**, and **Missing Data Handling** by suitable statistical and machine learning methods on the topic **Analyzing Creatinine Trajectory with Machine Learning Variable Selection**.

ONGOING RESEARCH AND READING PROJECTS

- **High Dimensional Statistical Learning | Dr. Debashis Paul** [Ongoing]
  - Currently studying the benign overfitting and it's conditions in various kinds of linear models.
- **Network Modeling on Poverty Data | Dr. Kiranmoy Das** [Ongoing]
  - Studied different subset selection methods (Forward Selection. Backward Elimination, Stepwise Regression, Best Subset Selection) and shrinkage methods(Ridge Regression, LASSO, Adaptive LASSO, Group LASSO, Fused LASSO)
  - Aiming towards fitting a network to identify key features behind multidimensional development of different countries.

SELECTED CLASS PROJECTS

- **Chronic Kidney Disease Progression in ADPKD** [Apr 2026]
  - We conducted a single-centre **Retrospective Longitudinal Cohort Study** of 193 patients with ADPKD followed across up to 26 outpatient visits.
- **Comparative Study of Maternal Health & Fertility Determinants** [Oct 2025]
  - Modeled **Early Childbearing and Parity Progression** using NFHS unit-level data for Bihar and Kerala via **Weighted Logistic and Quasi-Poisson Regression**.
- **Bayesian Mapping of Socioeconomic Health Determinants** [Nov 2025]

- Developed a **Unit-Level Bayesian Hierarchical Model** to estimate district-level household consumption—a primary proxy for nutritional security, using NSSO and Census data.

- **Machine Learning in Oncology: Breast Cancer Diagnostics** [Apr 2025]
  - Evaluated **LDA, QDA, and FDA** on tumor classification data; analyzed the stability of discriminant directions in high-dimensional medical settings.
- **Informatics: Computational Typography for OCR Optimization** [Apr 2023]
  - Designed character glyphs optimized for high-accuracy **Optical Character Recognition (OCR)**, focusing on automating medical record digitization.
- **Modeling Agricultural Productivity for Food Security** [Nov 2024]
  - Conducted predictive modeling of Indian crop production using **Multiple Linear Regression** to assess regional variations in agricultural output.
  - Analyzed environmental and regional covariates to identify structural drivers of food availability.

TECHNICAL SKILLS

- **Statistical Computing**: **R (Tidyverse, Survey Package, lme4, survival, geeglm)**, Python (NumPy, Matplotlib, Pandas, Scikit-learn).
- **Theoretical Foundations**:Inference, Linear Models, Asymptotic Theory, Multivariate Analysis.
- **Tools**: LaTeX, Git, Linux/Unix Shell.

RELEVANT COURSEWORK

- **Algorithms**: Design and Analysis of Algorithms, Numerical Analysis, Programming & Data Structures.
- **Core Theory**: Linear Statistical Models , Parametric Inference, Non-Parametric Inference, Design of Experiments, Stochastic Processes.
- **Foundations**: Real Analysis I/II/III, Vectors and Matrices I/II, Probability I/II/III.

LEADERSHIP & EXTRACURRICULARS

- **Class Representative**: Elected primary liaison between B.Stat batch and Faculty; managed academic scheduling.
- **Data Insights**: Actively analyzes international cricket and football tactics using domestic and global performance metrics.
- **Cultural**: Organized the 2024 Fresher's Welcome; managed logistics and venue coordination at ISI.